ONE-PAGE BRIEF

Post-quantum cryptography review: tachyon @ 6cdaebccf147

Client: Tachyon (Zcash) — public repository (passive review) · Source: github.com/tachyon-zcash/tachyon, commit `6cdaebccf147` · Reviewed: 2026-10-29 23:54 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **0** High **4** Medium
1 Low **2** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium QSI-2026-07445	**Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (1 file(s))** preliminary What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...
Medium QSI-2026-07443	**Quantum-vulnerable RedDSA: role not determined (6 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for...
Medium QSI-2026-07444	**Quantum-vulnerable RedDSA: authentication and identity (certificates, tokens, device attestation)...** preliminary What to do: Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them.
Medium QSI-2026-07442	**Non-cryptographic RNG near security-sensitive identifiers in book/mermaid.min.js** preliminary What to do: Use a CSPRNG (secrets, crypto.getRandomValues, crypto/rand).

1 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium threshold signing and MPC protocols	Pedersen commitment	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the...
Medium authentication and identity (certificates, tokens, device attestation)	RedDSA	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
Medium role not determined	RedDSA	Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 1; each needs checking against the code.

Static review of source code and manifests; files analysed: 58. No requests were sent to the owner's systems.

Full report, content SHA-256: `b4a5fab2bdec6a0f10dffb9b2bac126760c1defa80e6e0498eabf8830d661237`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.